

Week 3: Newton's Laws and Forces – Tutorial Sheet

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Use the Master Procedure: identify, isolate, draw the free-body diagram, choose axes, predict, write $\sum F = ma$ by components, solve, evaluate. Use $g = 9.80 \text{ m/s}^2$ and treat ropes and pulleys as ideal unless stated. Problem A1 was moved out of the lecture and is solved in full; the topics in section B were moved from Contact 3. Final answers to the rest are at the end.

A. Moved from the lecture

A1. A person in a lift

A 70.0 kg person stands on a scale in a lift. Find the scale reading n when the lift (a) moves upward at constant velocity, (b) accelerates upward at 1.5 m/s^2 , (c) accelerates downward at 1.5 m/s^2 , (d) is in free fall after the cable breaks.

Solution. FBD of the person: $\vec{n}$ up (scale on person), $\vec{w}$ down (Earth on person). Take $+y$ upward: $\sum F_y = n - mg = ma_y$, so $n = m(g + a_y)$. (a) $a_y = 0$: $n = 70.0(9.80) = 686 \text{ N}$, equal to the weight. (b) $a_y = +1.5$: $n = 70.0(11.3) = 791 \text{ N}$, larger than mg . (c) $a_y = -1.5$: $n = 70.0(8.3) = 581 \text{ N}$, smaller than mg . (d) $a_y = -g$: $n = 0$ ("weightless"). *Check:* the scale reading is the normal force, not the weight; the weight mg is the same in all four cases.

B. Practice

B1. Net force

Three forces act on an object: 100 N along $+x$, 60 N along $+y$ and 80 N at 225° from $+x$. Find the magnitude and direction of the net force, and the acceleration of a 5.0 kg object.

B2. Atwood machine

Two blocks of mass 5.0 kg and 3.0 kg hang from the ends of a light rope over a frictionless pulley. Find the acceleration and the tension.

B3. Connected objects with friction

A 4.0 kg block on a table with $\mu_k = 0.25$ is joined by a rope over a pulley to a hanging 2.0 kg block. Find the acceleration and the tension.

B4. Incline: limit of static friction

A block rests on an incline with $\mu_s = 0.60$. What is the largest angle for which it does not slide?

B5. Pulling up an incline

A 20.0 kg crate is pulled at constant speed up a 25° incline by a rope parallel to the slope. The coefficient of kinetic friction is 0.20. Find the tension.

B6. Braking lorry

A box rests on a lorry bed with $\mu_s = 0.40$. What is the greatest deceleration of the lorry for which the box does not slide?

C. Circular dynamics: moved from the lecture

C1. Banked curve

A frictionless banked curve of radius 100 m is to be taken at 20.0 m/s. At what angle must it be banked?

C2. Conical pendulum

A small mass hangs on a 1.00 m string and moves in a horizontal circle with the string at 30° from the vertical. Find its angular speed and the tension for $m = 0.50$ kg.

C3. Vertical circle

A 0.50 kg ball is whirled on a string in a vertical circle of radius 0.80 m. At the top the speed is 4.0 m/s. Find the tension at the top, and the least speed at the top for which the string stays taut.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

E1. Which of the following statements is/are true?

- I. If the net force on a body is zero, the body must be at rest.
- II. The two forces of a Newton's third-law pair act on different bodies.
- III. The mass of an object is the same on the Moon as on the Earth, but its weight is not.

(A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E2. A 75.0 kg person stands on a scale in a lift that is moving upward and slowing down at 1.50 m/s^2 . The scale reads

(A) 112 N (B) 368 N (C) 622 N (D) 735 N (E) 848 N

Items E3 and E4 share this information: Two blocks of mass 6.0 kg and 2.0 kg hang from the ends of a light rope that passes over a frictionless, massless pulley. They are released from rest.

E3. The magnitude of the acceleration of the blocks is

- (A) 2.45 m/s^2
- (B) 4.90 m/s^2
- (C) 6.53 m/s^2
- (D) 9.80 m/s^2
- (E) 19.60 m/s^2

E4. The tension in the rope is

- (A) 9.8 N (B) 19.6 N (C) 24.5 N (D) 29.4 N (E) 58.8 N

E5. A parcel rests on the bed of a lorry. When the lorry accelerates forward at 2.40 m/s^2 the parcel is just about to slip backward. The coefficient of static friction between parcel and bed is

- (A) 0.245 (B) 0.324 (C) 0.490 (D) 0.735 (E) 0.980

E6. An 8.0 kg block slides down a 30° incline with $\mu_k = 0.20$. Its acceleration down the slope is

- (A) 0.98 m/s^2
- (B) 1.96 m/s^2
- (C) 2.94 m/s^2
- (D) 3.20 m/s^2
- (E) 4.90 m/s^2

E7. A car takes an unbanked circular curve of radius 150 m at 18.0 m/s without skidding. The least coefficient of static friction needed is

- (A) 0.110 (B) 0.220 (C) 0.294 (D) 0.441 (E) 2.16

E8. A passenger standing in a bus falls forward when the bus brakes suddenly. Which of the following statements is/are true?

- I. No forward force acts on the passenger; the lurch is a consequence of inertia.
- II. The passenger's upper body tends to keep the velocity it had before the bus began to brake.
- III. A forward "inertial force" must be drawn on the free-body diagram of the passenger.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E9. A 0.40 kg ball on a string moves in a vertical circle of radius 0.90 m . At the bottom its speed is 5.0 m/s . The tension in the string at the bottom is

- (A) 3.9 N (B) 6.1 N (C) 7.2 N (D) 11.1 N (E) 15.0 N

E10. Two horizontal forces, 30.0 N east and 40.0 N north, act on a 5.0 kg block on a smooth surface. The magnitude of its acceleration is

- (A) 10.0 m/s^2
- (B) 6.0 m/s^2
- (C) 7.0 m/s^2
- (D) 8.0 m/s^2
- (E) 14.0 m/s^2

Answers

- B1 43.5 N at 4.5° above $+x$; $a = 8.7 \text{ m/s}^2$
B2 $a = 2.45 \text{ m/s}^2$; $T = 36.8 \text{ N}$
B3 $a = 1.6 \text{ m/s}^2$; $T = 16.3 \text{ N}$
B4 $\tan \theta = \mu_s$, so $\theta = 31.0^\circ$
B5 $T = mg(\sin \theta + \mu_k \cos \theta) = 118 \text{ N}$
B6 $a = \mu_s g = 3.9 \text{ m/s}^2$
C1 $\tan \theta = v^2/(gR) = 0.408$, so $\theta = 22.2^\circ$
C2 $\omega = \sqrt{g/(L \cos \theta)} = 3.36 \text{ rad/s}$; $T = mg/\cos \theta = 5.7 \text{ N}$
C3 $T = mv^2/R - mg = 5.1 \text{ N}$; $v_{\min} = \sqrt{gR} = 2.8 \text{ m/s}$

Exam-style practice: answer key and brief solutions

Key: E1 D E2 C E3 B E4 D E5 A E6 D E7 B E8 B E9 E E10 A

E1 (D). I is false: zero net force gives zero acceleration, so the velocity is constant, which includes constant non-zero velocity. II is true: a third-law pair acts on two different bodies, so the forces never cancel in one free-body diagram. III is true: m is a property of the object, $w = mg$ depends on g . Answer: II and III only.

E2 (C). Slowing down while moving up means the acceleration is downward, $a_y = -1.50 \text{ m/s}^2$. With $+y$ upward, $n - mg = ma_y$, so $n = m(g + a_y) = 75.0(9.80 - 1.50) = 623 \text{ N}$, less than $mg = 735 \text{ N}$ as a downward acceleration predicts.

E3 (B). Write $\sum F = ma$ for each block with the same a (ideal rope): $m_1g - T = m_1a$ and $T - m_2g = m_2a$. Adding, $a = \frac{m_1 - m_2}{m_1 + m_2}g = \frac{4.0}{8.0}(9.80) = 4.90 \text{ m/s}^2$.

E4 (D). Use the acceleration of the previous item in the equation for the lighter block: $T = m_2(g + a) = 2.0(9.80 + 4.90) = 29.4 \text{ N}$. Check with the heavier block: $T = m_1(g - a) = 6.0(4.90) = 29.4 \text{ N}$. T lies between the two weights (19.6 N and 58.8 N).

E5 (A). The only horizontal force on the parcel is static friction from the bed, forward, and it is about to reach its limit: $\mu_s n = ma$ with $n = mg$. So $\mu_s = a/g = 2.40/9.80 = 0.245$. The mass cancels.

E6 (D). Axes along the slope. $\sum F_x = mg \sin \theta - \mu_k mg \cos \theta = ma$, so $a = g(\sin 30^\circ - 0.20 \cos 30^\circ) = 9.80(0.500 - 0.173) = 3.20 \text{ m/s}^2$. The mass cancels. It is smaller than the frictionless value 4.90 m/s^2 , as predicted.

E7 (B). Static friction supplies the centripetal force: $\mu_s mg \geq mv^2/r$, so $\mu_s \geq v^2/(rg) = \frac{18.0^2}{(150)(9.80)} = 0.220$. The mass cancels. The value 2.16 is the centripetal acceleration in m/s^2 , not a coefficient.

E8 (B). I is true: the only horizontal force is friction on the feet, directed backward. II is true (first law). III is false: a free-body diagram contains only real forces exerted by other objects, so the “inertial force” is not drawn. Answer: I and II only.

E9 (E). At the bottom the centre is above the ball: take $+y$ up. $T - mg = mv^2/r$, so $T = m(g + v^2/r) = 0.40(9.80 + 27.8) = 15.0 \text{ N}$. Larger than $mg = 3.9 \text{ N}$, as predicted. Using $T = mv^2/r$ alone omits the weight.

E10 (A). Add the forces as vectors: $F = \sqrt{30.0^2 + 40.0^2} = 50.0 \text{ N}$ (a 3-4-5 triangle), so $a = F/m = 10.0 \text{ m/s}^2$, directed 53° north of east. Adding the magnitudes gives 14 m/s^2 .